HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and trained a **gradient-boosted classifier** using scikit-learn on a dataset of approximately 280,000 transaction records to identify suspicious activity, achieving an **AUC of 0.86** on held-out data.
- Engineered **Python feature pipelines** on AWS Lambda and S3 to support downstream ML model scoring, processing around **3,500 events per second** during peak hours and replacing a less efficient scheduled batch job.
- Designed and implemented **REST APIs in Java Spring Boot** for inter-service data exchange, incorporating request validation and circuit-breaker patterns to enhance system resilience.
- Contributed to applied **machine learning research** informing product experiences by collecting, analyzing, and synthesizing data for model development.
- Wrote clean, efficient, and maintainable **Python code**, supported by appropriate documentation and unit tests for data processing and model integration.
- Collaborated with engineering and data teams to prototype, evaluate, and improve **ML-powered features**, contributing to discovery experiences.
- Containerized microservices using **Docker** and deployed them to an **AWS EKS cluster**, configuring Helm charts and resource limits for predictable rollouts.

SKILLS

Programming Languages – Python, Java, SQL, R

ML Frameworks & Libraries – scikit-learn, pandas, NumPy, TensorFlow, PyTorch

Cloud Platforms & Services – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes (EKS)

Databases & Data Warehousing – PostgreSQL, MySQL, Oracle, SQL

ML Concepts – Machine Learning, NLP, Ranking, Recommendations, Embeddings, Information Retrieval, MLOps

Big Data Technologies – Hadoop, Spark

PROJECTS

ML Model for Transaction Anomaly Detection

- Engineered a **Python feature pipeline** to extract behavioural signals from a transaction log of approximately 1.1 million records.
- Trained a **LightGBM classifier** on extracted features, tuning class weights to address class imbalance and achieving an **AUC of 0.91** and an F1 score of 0.46 on a held-out month.
- Packaged the trained model behind a **FastAPI inference endpoint** with input validation and basic feature-drift logging, achieving a p95 latency of 180ms.

Customer Onboarding for an Internal ML Platform

- Worked directly with two pilot customers to design and ship a guided onboarding flow for an **internal ML feature store**, including SDK examples and debug tooling.
- Built a **Python CLI** to validate customer feature definitions against the platform schema before submission, catching approximately **70% of schema errors** at write time.
- Documented common integration patterns and ran follow-up workshops with customer data teams to upskill their internal engineers on the ML platform.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)